StudySapiens Board MCQs

Chapter 2 — Bacteria and Viruses

Board-style MCQs · Questions first, answer key with brief reasons at the end · PECTAA 2025–26

Choose the best option. Each question carries one mark. The answer key with brief reasons is given at the end of this chapter.

1. Bacteria are best described as:

(a) Unicellular eukaryotes
(b) Acellular particles
(c) Unicellular prokaryotes
(d) Multicellular prokaryotes

2. The major component of the bacterial cell wall is:

(a) Peptidoglycan
(b) Silica
(c) Chitin
(d) Cellulose

3. Gram-positive bacteria have:

(a) A chitin wall
(b) A thick peptidoglycan layer
(c) No cell wall
(d) A thin peptidoglycan layer and outer membrane

4. The outer membrane of Gram-negative bacteria contains:

(a) Lipopolysaccharides
(b) Cellulose
(c) Peptidoglycan only
(d) Chitin

5. Porins in Gram-negative bacteria function as:

(a) Photosynthetic pigments
(b) Motility organs
(c) Genetic material
(d) Pores for specific molecules

6. The gelatinous layer outside some bacterial cell walls is called:

(a) Plasmid
(b) Nucleoid
(c) Capsule
(d) Mesosome

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7. Invaginations of the bacterial plasma membrane are called:

(a) Capsules
(b) Mesosomes
(c) Ribosomes
(d) Plasmids

8. Bacterial ribosomes are:

(a) 80S
(b) 70S
(c) 40S
(d) 60S

9. The bacterial chromosome is usually:

(a) Circular and without histones
(b) Linear with histones
(c) RNA only
(d) Enclosed in a nuclear envelope

10. Plasmids are useful in genetic engineering because they:

(a) Produce ATP
(b) Are viral coats
(c) Form the cell wall
(d) Can carry selected genes into bacteria

11. Endospores are formed mainly for:

(a) Survival in harsh conditions
(b) Photosynthesis
(c) Reproduction
(d) Conjugation

12. The process of endospore formation is called:

(a) Budding
(b) Sporulation
(c) Fragmentation
(d) Binary fission

13. During sporulation, the vegetative cell ultimately:

(a) Becomes a virus
(b) Breaks and releases the endospore
(c) Forms a capsule
(d) Divides into two endospores

14. When favourable conditions return, an endospore:

(a) Forms a capsule only
(b) Remains dormant forever
(c) Germinates into a vegetative cell
(d) Becomes a plasmid

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15. Movement of a bacterial population together by flagella is called:

(a) Twitching
(b) Swarming
(c) Gliding
(d) Sliding

16. Twitching motility is mediated by:

(a) Pili
(b) Flagella
(c) Endospores
(d) Capsule

17. Gliding motility is helped by secretion of:

(a) Slimy substance
(b) Spores
(c) Antibiotics
(d) Peptidoglycan

18. Spirochaetes move using:

(a) Capsules
(b) Axial filaments
(c) Pseudopodia
(d) Cilia

19. Bacteria without flagella are called:

(a) Lophotrichous
(b) Monotrichous
(c) Peritrichous
(d) Atrichous

20. A single polar flagellum is characteristic of:

(a) Monotrichous bacteria
(b) Lophotrichous bacteria
(c) Amphitrichous bacteria
(d) Peritrichous bacteria

21. Flagella distributed all over the cell surface are:

(a) Peritrichous
(b) Lophotrichous
(c) Monotrichous
(d) Atrichous

22. A tuft of flagella at one pole is called:

(a) Lophotrichous
(b) Peritrichous
(c) Monotrichous
(d) Amphitrichous

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23. The bacterial flagellum is composed mainly of:

(a) Flagellin
(b) Actin
(c) Keratin
(d) Tubulin

24. The basal body of a Gram-positive bacterial flagellum has:

(a) Four pairs of rings
(b) Two pairs of rings
(c) No rings
(d) One pair of rings

25. Which group of bacteria lacks a cell wall?

(a) Cyanobacteria
(b) Spirochaetes
(c) Actinomycetes
(d) Mycoplasmas

26. Mycoplasmas are resistant to penicillin because they:

(a) Produce endospores
(b) Lack a cell wall
(c) Are viruses
(d) Have thick peptidoglycan

27. Cyanobacteria are important because they:

(a) Lack pigments
(b) Cause malaria
(c) Are all pathogenic
(d) Increase atmospheric oxygen and fix nitrogen

28. Nitrogen fixation in many cyanobacteria occurs in:

(a) Endospores
(b) Capsules
(c) Heterocysts
(d) Plasmids

29. Which bacteria cause syphilis?

(a) Nitrosomonas
(b) Lactobacillus
(c) Rhizobium
(d) Treponema

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30. Many antibiotics such as tetracycline and erythromycin were originally derived from:

(a) Mycoplasmas
(b) Cyanobacteria
(c) Viruses
(d) Actinomycetes

31. Azotobacter is a:

(a) Virus
(b) Free-living nitrogen-fixing bacterium
(c) Protozoan
(d) Fungus

32. Nitrosomonas and Nitrobacter are important in:

(a) The nitrogen cycle
(b) Photosynthesis only
(c) Viral replication
(d) Alcoholic fermentation

33. Bacteria used to clean oil spills are useful because they:

(a) Are photosynthetic
(b) Digest hydrocarbons
(c) Produce oxygen
(d) Form endospores only

34. Bacillus thuringiensis is used as a:

(a) Virus
(b) Fungus
(c) Biopesticide
(d) Human pathogen

35. Bacteria are used in genetic engineering to produce:

(a) Only alcohol
(b) Only antibiotics
(c) Only vinegar
(d) Insulin and other therapeutic proteins

36. The mixture of microorganisms regularly found at an anatomical site is called:

(a) Capsule
(b) Normal flora
(c) Endospore
(d) Pathogen

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37. Normal flora of the human intestine help by producing:

(a) Antibodies only
(b) Vitamin C only
(c) Insulin
(d) Vitamin K and B12

38. Normal flora prevent pathogen colonization by:

(a) Producing oxygen
(b) Competing for attachment sites and nutrients
(c) Forming endospores
(d) Killing all viruses

39. Viruses can be seen only with a/an:

(a) Light microscope
(b) Hand lens
(c) Electron microscope
(d) Telescope

40. The genetic material of a virus may be:

(a) DNA or RNA
(b) Protein only
(c) RNA only
(d) DNA only

41. The protein subunits of a capsid are called:

(a) Amino acids
(b) Nucleotides
(c) Capsomeres
(d) Porins

42. The core plus capsid of a virus is called the:

(a) Envelope
(b) Nucleocapsid
(c) Prion
(d) Virion only

43. Non-enveloped viruses are called:

(a) Viroids
(b) Naked viruses
(c) Retroviruses
(d) Bacteriophages

44. The viral envelope is derived from:

(a) The nucleoid
(b) The viral capsid
(c) The cell wall
(d) The host cell membrane

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45. Bacteriophages attack:

(a) Fungi only
(b) Bacteria
(c) Animals only
(d) Plants only

46. The head of a T4 phage contains:

(a) Protein only
(b) RNA only
(c) Double-stranded DNA
(d) Lipids only

47. Tail fibres of a bacteriophage help in:

(a) Photosynthesis
(b) Protein synthesis
(c) DNA replication
(d) Attachment to the bacterial wall

48. HIV belongs to the group:

(a) Prions
(b) Viroids
(c) Bacteriophages
(d) Retroviruses

49. The enzyme reverse transcriptase in HIV:

(a) Makes DNA from RNA
(b) Makes RNA from DNA
(c) Digests protein
(d) Forms the capsid

50. Glycoprotein spikes of HIV help the virus:

(a) Fix nitrogen
(b) Replicate DNA
(c) Form endospores
(d) Identify and bind to target cells

51. Prions are composed of:

(a) Protein only
(b) DNA and protein
(c) DNA only
(d) RNA only

52. Viroids are composed of:

(a) Protein only
(b) Circular RNA only
(c) Lipids only
(d) DNA and RNA

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53. Which of the following is NOT a characteristic of viruses?

(a) They have their own ribosomes
(b) They contain nucleic acid
(c) They have a protein coat
(d) They require a host to replicate

54. The size range of most viruses is approximately:

(a) 20–250 nm
(b) 1–5 mm
(c) 100–500 µm
(d) 1–10 µm

55. Which bacteria are commonly used in fermented foods such as yogurt?

(a) Mycobacterium
(b) Vibrio only
(c) Treponema
(d) Lactobacillus

56. Decomposer bacteria are important because they:

(a) Produce oxygen only
(b) Form viruses
(c) Cause all diseases
(d) Recycle nutrients from dead organic matter

57. Denitrification is carried out by bacteria such as:

(a) Pseudomonas
(b) Lactobacillus
(c) Rhizobium
(d) Streptococcus

58. The fossil record suggests prokaryotes were abundant about:

(a) 1 million years ago
(b) 3.5 billion years ago
(c) 10,000 years ago
(d) 500 million years ago only

59. Which of the following is a plant disease caused by spiroplasmas?

(a) Malaria
(b) Syphilis
(c) Lethal yellowing of coconuts
(d) Tuberculosis

60. Mycobacterium tuberculosis causes:

(a) Leprosy
(b) Pneumonia only
(c) Syphilis
(d) Tuberculosis

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61. Mycobacterium leprae causes:

(a) Leprosy
(b) Cholera
(c) Typhoid
(d) Tuberculosis

62. The secondary function of bacterial flagella is to:

(a) Synthesize protein
(b) Form endospores
(c) Detect and respond to chemical signals
(d) Store fat

63. Which motility type is due to random particle movement rather than cellular structures?

(a) Gliding
(b) Twitching
(c) Brownian movement
(d) Swarming

64. In Gram-negative bacteria, the periplasmic space is:

(a) Absent
(b) More prominent than in Gram-positive bacteria
(c) Part of the capsule
(d) Inside the nucleoid

65. The capsule of bacteria mainly helps in:

(a) Attachment and protection
(b) DNA replication
(c) Protein synthesis
(d) Photosynthesis

66. Which of the following is a beneficial use of bacteria in the environment?

(a) Forming viral capsids
(b) Producing toxins only
(c) Causing food spoilage only
(d) Bioremediation

67. Normal flora stimulate the immune system by:

(a) Acting as antigens and inducing cross-reactive antibodies
(b) Forming endospores
(c) Producing insulin
(d) Killing all pathogens directly

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68. Which of the following is true about viruses?

(a) They reproduce by binary fission
(b) They are prokaryotic cells
(c) They are obligate intracellular parasites
(d) They have both DNA and RNA

69. Which of the following is used as a carrier in genetic engineering to transfer genes into bacteria?

(a) Endospore
(b) Mesosome
(c) Bacteriophage
(d) Capsule

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ANSWER KEY WITH BRIEF REASONS

1. (c) Unicellular prokaryotes — Bacteria lack a true nucleus and membrane-bound organelles.

2. (a) Peptidoglycan — Bacterial walls are made of peptidoglycan/murein.

3. (b) A thick peptidoglycan layer — Gram-positive walls are thick in peptidoglycan and retain crystal violet.

4. (a) Lipopolysaccharides — Gram-negative bacteria have an LPS-rich outer membrane.

5. (d) Pores for specific molecules — Porins allow passage of selected molecules through the outer membrane.

6. (c) Capsule — Capsules give sticky character and protection.

7. (b) Mesosomes — Mesosomes are membrane invaginations implicated in DNA replication/division.

8. (b) 70S — Bacterial ribosomes are 70S (50S + 30S).

9. (a) Circular and without histones — Bacterial DNA is a circular molecule in the nucleoid.

10. (d) Can carry selected genes into bacteria — Plasmids are self-replicating vectors.

11. (a) Survival in harsh conditions — Endospores are dormant, resistant survival structures.

12. (b) Sporulation — Sporulation produces a resistant endospore.

13. (b) Breaks and releases the endospore — The mature endospore is released when the vegetative cell breaks.

14. (c) Germinates into a vegetative cell — Germination restores active bacterial life.

15. (b) Swarming — Swarming is coordinated flagellar movement.

16. (a) Pili — Pili attach and retract to pull the cell forward.

17. (a) Slimy substance — Slime allows smooth movement over surfaces.

18. (b) Axial filaments — Axial filaments lie in the periplasmic space and produce flexing/spinning.

19. (d) Atrichous — Atrichous bacteria lack flagella.

20. (a) Monotrichous bacteria — Monotrichous means one polar flagellum.

36. (b) Normal flora — Normal flora colonize body surfaces.

37. (d) Vitamin K and B12 — Enteric bacteria synthesize vitamin K and B12.

38. (b) Competing for attachment sites and nutrients — Competition reduces pathogen establishment.

39. (c) Electron microscope — Viruses are extremely small, usually 20–250 nm.

40. (a) DNA or RNA — Viruses contain either DNA or RNA, not both.

41. (c) Capsomeres — Capsomeres make up the capsid.

42. (b) Nucleocapsid — Nucleocapsid = genome + capsid.

43. (b) Naked viruses — Naked viruses lack a lipid envelope.

44. (d) The host cell membrane — Envelopes are host-derived lipid membranes.

45. (b) Bacteria — Bacteriophages are bacterial viruses.

46. (c) Double-stranded DNA — T4 phage has a dsDNA genome in its head.

47. (d) Attachment to the bacterial wall — Tail fibres recognize and attach to the host.

48. (d) Retroviruses — HIV is an RNA retrovirus.

49. (a) Makes DNA from RNA — Reverse transcriptase synthesizes DNA from an RNA template.

50. (d) Identify and bind to target cells — Spikes mediate host-cell recognition and binding.

51. (a) Protein only — Prions are infectious proteins.

52. (b) Circular RNA only — Viroids are naked circular RNA molecules.

53. (a) They have their own ribosomes — Viruses lack ribosomes and metabolism.

54. (a) 20–250 nm — Viruses are much smaller than bacteria.

55. (d) Lactobacillus — Lactobacillus is used in dairy fermentation.

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21. (a) Peritrichous — Peritrichous flagella surround the whole cell.

22. (a) Lophotrichous — Lophotrichous means a polar tuft.

23. (a) Flagellin — Bacterial flagella are made of flagellin, not microtubules.

24. (d) One pair of rings — Gram-positive basal bodies have one pair of rings.

25. (d) Mycoplasmas — Mycoplasmas and spiroplasmas lack cell walls.

26. (b) Lack a cell wall — Penicillin targets cell wall synthesis.

27. (d) Increase atmospheric oxygen and fix nitrogen — Cyanobacteria photosynthesize and many fix nitrogen in heterocysts.

28. (c) Heterocysts — Heterocysts are specialized nitrogen-fixing cells.

29. (d) Treponema — Treponema pallidum is a spirochaete causing syphilis.

30. (d) Actinomycetes — Actinomycetes are a major source of antibiotics.

31. (b) Free-living nitrogen-fixing bacterium — Azotobacter fixes atmospheric nitrogen in soil/water.

32. (a) The nitrogen cycle — They oxidize nitrogen compounds during nitrification.

33. (b) Digest hydrocarbons — Some bacteria degrade petroleum hydrocarbons.

34. (c) Biopesticide — Bt is used in biological pest control.

35. (d) Insulin and other therapeutic proteins — Human genes can be expressed in bacteria to make proteins.

56. (d) Recycle nutrients from dead organic matter — Decomposers complete biogeochemical cycles.

57. (a) Pseudomonas — Some Pseudomonas species reduce nitrates to nitrogen gas.

58. (b) 3.5 billion years ago — Prokaryotes appeared very early in Earth history.

59. (c) Lethal yellowing of coconuts — Spiroplasmas cause lethal yellowing in palms.

60. (d) Tuberculosis — M. tuberculosis is the causative agent of TB.

61. (a) Leprosy — M. leprae causes leprosy.

62. (c) Detect and respond to chemical signals — Flagella also help in chemotaxis.

63. (c) Brownian movement — Brownian movement is passive/random.

64. (b) More prominent than in Gram-positive bacteria — Gram-negative bacteria have a larger periplasmic space.

65. (a) Attachment and protection — Capsules are sticky and protective.

66. (d) Bioremediation — Bacteria can degrade pollutants.

67. (a) Acting as antigens and inducing cross-reactive antibodies — Low-level immune responses to flora can cross-react with pathogens.

68. (c) They are obligate intracellular parasites — Viruses replicate only inside host cells.

69. (c) Bacteriophage — Phage DNA can carry a gene of interest into a bacterial cell.

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